

Concurrency and Distributed Systems

Week 10 – Part 1: Network Laws

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Part 1 Roadmap

- 1 Phase 3
- 2 Three Laws
- 3 Demo

The Phase 3 Shift

Before

Threads, locks, queues, futures, and runtime limits inside one machine.

Now

Messages cross a network boundary. There is no shared memory.

Three Laws of Networks

- ① Latency is never zero.
- ② Bandwidth is never infinite.
- ③ Failures are partial and invisible.

Design implication

Treat every network call as waiting plus uncertainty.

Retry Amplifies Load

- A retry is extra work during an already bad condition.
- Timeout without backoff can create a request storm.
- Backpressure now crosses process boundaries.

Run

```
.\run-demo.ps1 01
```